in the hardware industry was what hampered progress. The present wave of ANN is fuelled by unprecedented growth in the hardware industry in the form of ultrafast CPUs and GPUs with multiple cores, all available at an affordable cost.

Deep learning

The ANN has a layered approach. It has one input layer and one output layer. There may be one or more hidden layers. When the constructed network becomes deep (with a higher number of hidden layers), it is called a deep network or deep learning network. The popularity of deep learning can be understood from the fact that leading IT companies like Google, Facebook, Microsoft and Baidu have invested in it in domains like speech, image and behaviour modelling. Deep learning has various characteristics as listed below:

- In the traditional approach, the features of machine learning or shallow learning have to be identified by humans whereas in deep learning, the features are not human constructed.
- Deep learning involves ANN with a greater number of layers.
- Deep learning handles end-to-end compositional models. The hierarchy of representations with different data is handled. For example, with speech, the hierarchical representation is: *Audio -> Band -> Phone -> Word*. For vision, it is: *Pixel -> Motif -> Part -> Object*.

There are plenty of resources available on the Web to understand deep learning. However, at times it leads to the problem of plenty. The lecture delivered by Dr Andrew Ng at the GPU Technology Conference 2015 is one of the best to start with, in order to get a clear understanding of deep learning and its potential applications (*http://www.ustream.tv/recorded/60113824*).

Deep learning domains

Deep learning has applications in various verticals. The three popular verticals are shown in Figure 1.

In the image domain, there are applications like classification, identification, captioning, matching, etc. The speech domain has applications like real time recognition under noisy environments. The behaviour domain has applications like trend analysis, security aspects, etc. Deep learning research studies are conducted in domains such as natural language processing, etc.

Deep learning libraries

There are many libraries that help to make the deep learning implementations quicker (Figure 2), as they make a layer of abstraction over the underlying complex mathematical representations.

Some of these libraries may turn out to be the trendsetters in 2016 and the years ahead. This article covers five of these libraries and their extensions. As deep learning itself is a bleeding edge technology, the field is in a dynamic stage. The libraries chosen for this article are representative in nature.

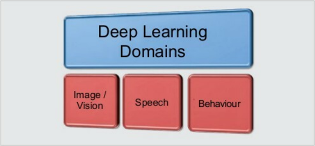

Figure 1: Deep learning domains

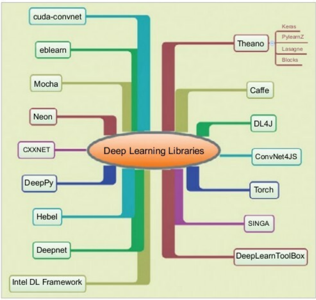

Figure 2: Deep learning libraries

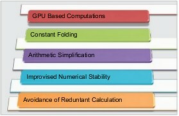

Figure 3: Theano features

Theano

Theano is a Python library for defining, evaluating and optimising mathematical expressions. As deep learning involves tons of matrix/vector manipulation, Theano is employed for those tasks efficiently. It enables developers to attain the speed of execution that is possible only with direct C implementations. The greater benefit with Theano is that it can harness resources of both the CPU and the GPU.

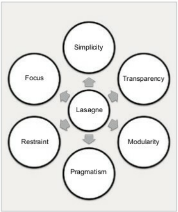

Figure 4: Caffe image classification demo